

MUHAMMAD HAZIQ IMRAN

AI Engineer

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PROFESSIONAL SUMMARY

BS Artificial Intelligence graduate with hands-on experience building and deploying production-ready Generative AI systems, LLM-powered agents, RAG pipelines, and deep learning models. Proficient in Python, LangChain, LangGraph, OpenAI API (GPT-4), HuggingFace Transformers, and vector databases (Pinecone). Experienced in constructing multi-agent architectures, agentic workflows, conversational AI, computer vision systems, NLP solutions, and AI-integrated REST APIs (FastAPI/Flask). Skilled in data science, time-series analysis, signal processing, and ML model evaluation. Passionate about prompt engineering, retrieval-augmented generation, and applied machine learning.

EDUCATION

BS Artificial Intelligence | Khwaja Fareed University of Engineering & IT, Rahim Yar Khan 2022 – 2026

Graduated 2026 | Coursework: Machine Learning, Deep Learning, NLP, Computer Vision, Reinforcement Learning, Database Systems, Programming for Artificial Intelligence

TECHNICAL SKILLS

Languages: Python

Generative AI & LLMs: LangChain, LangGraph, OpenAI API, HuggingFace Transformers, Groq, Prompt Engineering, RAG (Retrieval-Augmented Generation) Pipelines, Agentic Workflows, Multi-Agent Systems, Conversational AI, LLM Tool Calling, Function Calling, LLM Integration

Machine Learning & Deep Learning: TensorFlow, Keras, PyTorch, Scikit-learn, CNN, LSTM, RNN, Transfer Learning, Fine-tuning, Model Evaluation, Hyperparameter Tuning, Feature Engineering, Text Classification, Sentiment Analysis

Computer Vision: OpenCV, YOLOv8, MediaPipe, Pose Estimation, Object Detection, Real-Time Object Tracking, Keypoint Detection, Image Processing

NLP & Data Science: NLP, Semantic Search, Text Embeddings, Named Entity Recognition, Summarization, NumPy, Pandas, Data Preprocessing, EDA, Time-Series Analysis, Signal Processing, FAISS

Databases & Vector Stores: Pinecone, MongoDB, SQLite, Firebase, SQLAlchemy, ChromaDB

APIs & Frameworks: FastAPI, Flask, Streamlit, REST API, WhatsApp API, Gmail API

Tools & DevOps: Git, GitHub, Docker, Selenium, BeautifulSoup, Web Scraping

CERTIFICATIONS

- Career Essentials in Generative AI — Microsoft & LinkedIn Learning
- AI Advanced Course — Xeven Solutions

WORK EXPERIENCE

AI/ML Engineer Intern | Ezitech Institute Jun 2025 – Sep 2025

- Fine-tuned a BERT-based Fake News Detection model using HuggingFace Transformers, achieving ~93% classification accuracy on a labeled NLP dataset of 5,000+ news articles.
- Engineered a Phishing Website Detection classifier using Scikit-learn ML pipeline with feature extraction and model evaluation; attained ~91% accuracy with F1-Score of 0.90 and ROC-AUC of 0.95.
- Constructed an end-to-end Comic Story Generator pipeline combining Groq LLM for narrative generation with Coqui TTS for automated voice narration, reducing manual content creation time by ~70%.
- Completed 3 end-to-end ML projects covering NLP, classification, and Generative AI, applying industry-standard preprocessing, hyperparameter tuning, and performance benchmarking with Python, Pandas, and NumPy.

FINAL YEAR PROJECT

ShopGen AI — WhatsApp-Integrated Conversational Shopping Agent | Khwaja Fareed University of Engineering & IT 2025 – 2026

- Architected a production-ready multi-agent AI system on WhatsApp using LangGraph agentic workflows for autonomous product discovery, real-time price comparison, personalized fashion recommendations, and conversational commerce.
- Implemented stateful agent memory with persistent session context via LangChain and LangGraph, enabling coherent multi-turn conversations and long-term user preference tracking across 100+ interactions.
- Integrated Pinecone vector database for semantic product search via dense embeddings and Firebase Realtime Database for inventory and session management supporting concurrent users.

- Built a multimodal I/O pipeline supporting text and voice (STT via OpenAI Whisper / TTS) over WhatsApp via Whapi REST API; reduced query response latency to under 3 seconds.
- Stack: Python, LangGraph, LangChain, OpenAI GPT-4o Mini, OpenAI Whisper, Whapi, Pinecone, Firebase, FastAPI, SerpAPI

PROJECTS

AI Scholarship Assistant — AI Scholarship Guidance Platform (Team Project) | *Python, Flask, RAG, LLMs, OpenAI GPT, NLP, SQLite, SQLAlchemy, Pinecone, Tailwind CSS, ReportLab*

- Co-built a full-stack Generative AI scholarship platform featuring 12+ integrated AI tools for students across 25+ countries, with RAG-based conversational AI and LLM-powered personalization.
- Created a RAG chatbot over a curated knowledge base of 120+ global scholarships for intelligent semantic search, Q&A, and opportunity matching via text embeddings and vector retrieval.
- Architected an LLM-powered AI Mentor for personalized scholarship recommendations and an ATS Resume Checker with PDF parsing, keyword extraction, and compatibility scoring.

MindBridge AI — Conversational AI Mental Wellness Companion | *Python, LLMs, NLP, FastAPI, Pinecone, LangChain, Prompt Engineering*

- Developed an empathetic conversational AI assistant using LLMs and advanced prompt engineering for context-aware emotional support, maintaining continuity across multiple sessions via RAG-based memory retrieval.
- Engineered automatic session summarization and adaptive user profiling with LangChain memory modules, personalizing guidance across 50+ simulated user interactions.

ProfReach — AI Professor Outreach Automation (Ongoing) | *Python, LangGraph, LLMs, NLP, Selenium, BeautifulSoup, Gmail API, FastAPI, Prompt Engineering*

- Developing an agentic AI automation pipeline using LangGraph, LLMs and web scraping (Selenium, BeautifulSoup) to identify professors, analyze 100+ research profiles, and generate hyper-personalized outreach emails.
- Applying semantic similarity matching via text embeddings to align student research interests with professor publications; automated Gmail API delivery and response tracking, cutting manual outreach time by ~80%.

MMA Vision — Multi-Person Martial Arts Pose Detection | *Python, OpenCV, YOLOv8, MediaPipe, Deep Learning, Computer Vision*

- Designed a real-time computer vision system using YOLOv8 object detection and MediaPipe pose estimation to track 15+ body keypoints per fighter across simultaneous multi-person video streams.
- Extracted joint angles and motion patterns for activity recognition and sports analytics; achieved real-time inference at 25+ FPS on standard hardware.

Epileptic Seizure Prediction — EEG Signal Analysis | *Python, Scikit-learn, Decision Tree, Random Forest, XGBoost, NumPy, Pandas, Signal Processing*

- Created a healthcare AI system using Decision Tree, Random Forest, and XGBoost models for early epileptic seizure prediction from EEG signals, achieving ~88% accuracy on a benchmark EEG dataset.
- Built end-to-end EEG signal preprocessing pipeline — noise reduction, normalization, feature extraction, and time-series segmentation — and evaluated with F1-Score, ROC-AUC, and cross-validation.